

KESHAV MAHAVIDYALAYA

(UNIVERSITY OF DELHI)

Practical List

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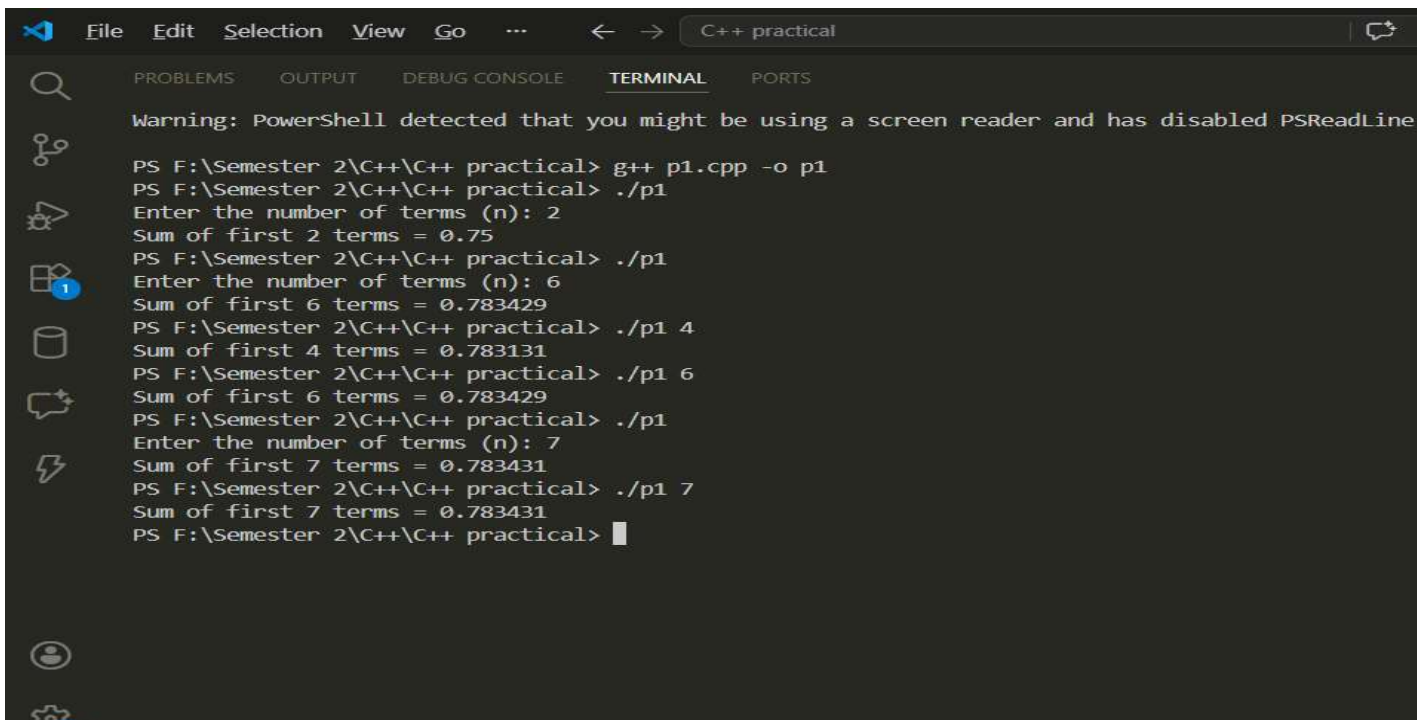
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""(P1) Write a program to compute the sum of the first n terms of the following series: $S = 1 - 1/2^2 + 1/3^3 - 1/4^4 + 1/5^5 - \dots 1/n^n$. The number of terms n is to be taken from the user through the command line. If the command line argument is not found then prompt the user to enter the value of n.""

```
#include <iostream>
#include <cmath>
using namespace std;
int main(int argc, char* argv[]) {
    int n;
    if (argc > 1) {
        n = stoi(argv[1]);
    }
    else {
        cout << "Enter the number of terms (n): ";
        cin >> n;
    }
    double sum = 0.0;

    for (int i = 1; i <= n; i++) {
        double term = 1.0 / pow(i, i);
        if (i % 2 == 0) {
            sum -= term;
        } else {
            sum += term;
        }
    }
    cout << "Sum of first " << n << " terms = " << sum << endl;
    return 0;
}
```



The screenshot shows a Visual Studio Code terminal window with the title "C++ practical". The terminal output shows the program being compiled and executed multiple times with different values of n. The output for each execution is as follows:

```
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine
PS F:\Semester 2\C++\C++ practical> g++ p1.cpp -o p1
PS F:\Semester 2\C++\C++ practical> ./p1
Enter the number of terms (n): 2
Sum of first 2 terms = 0.75
PS F:\Semester 2\C++\C++ practical> ./p1
Enter the number of terms (n): 6
Sum of first 6 terms = 0.783429
PS F:\Semester 2\C++\C++ practical> ./p1 4
Sum of first 4 terms = 0.783131
PS F:\Semester 2\C++\C++ practical> ./p1 6
Sum of first 6 terms = 0.783429
PS F:\Semester 2\C++\C++ practical> ./p1
Enter the number of terms (n): 7
Sum of first 7 terms = 0.783431
PS F:\Semester 2\C++\C++ practical> ./p1 7
Sum of first 7 terms = 0.783431
PS F:\Semester 2\C++\C++ practical>
```

#(P2): Write a program to remove duplicates from an array.

```
#include <iostream>
using namespace std;
int main(){
    int n;

    cout<<"Enter the no. of elements:";
    cin>>n;

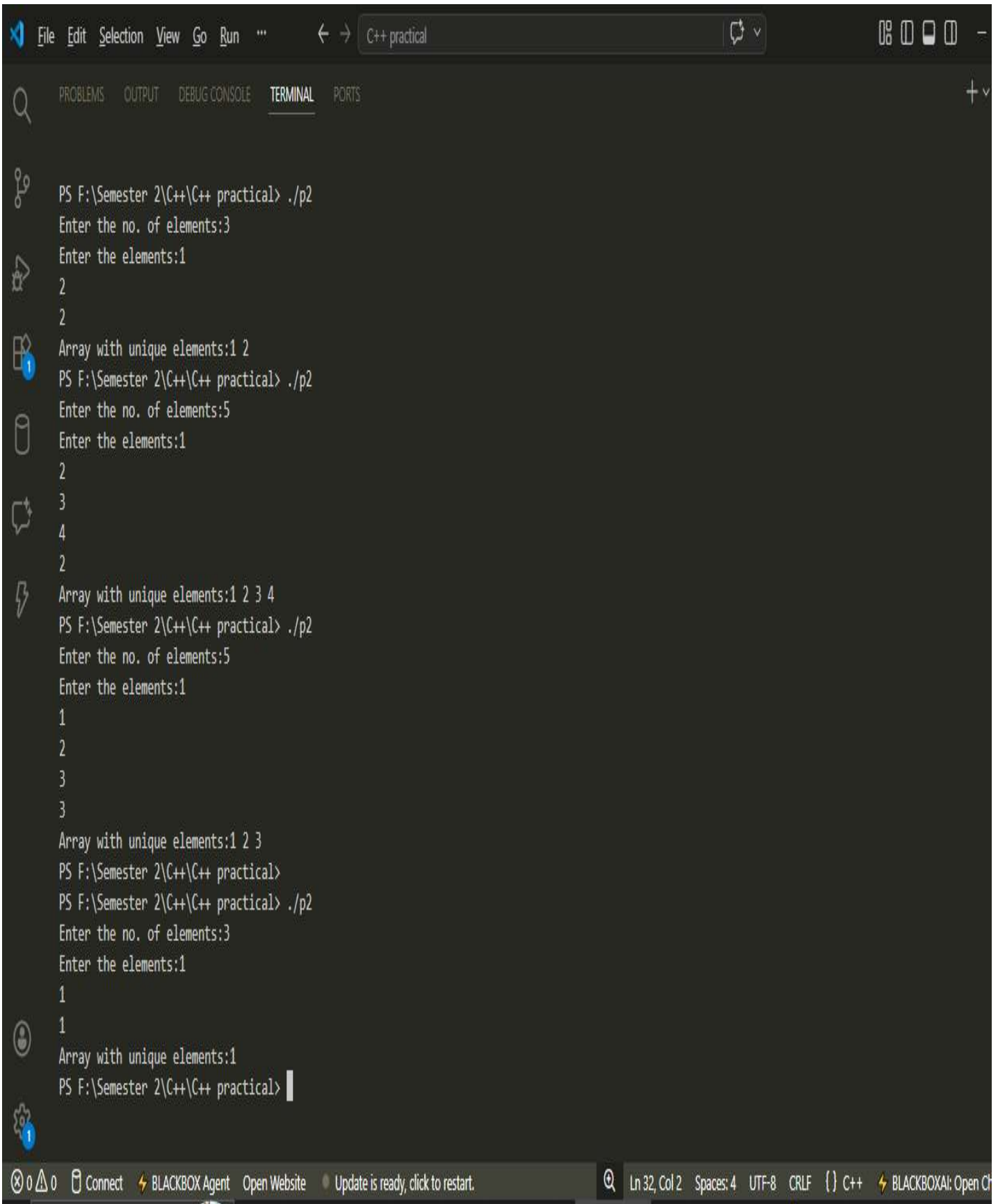
    int arr[n];
    cout<<"Enter the elements:";
    for (int i=0; i<n; i++){
        cin>>arr[i];
    }

    int arr2[n];
    int m=0;
    for (int i=0; i<n; i++){
        bool found=false;
        for (int j=0; j<m; j++){
            if(arr[i]==arr2[j]){
                found=true;
                break;
            }
        }

        if(!found){
            arr2[m]=arr[i];
            m++;
        }
    }

    cout<<"Array with unique elements:";
    for (int i=0; i<m; i++){
        cout<<arr2[i]<<" ";
    }

    return 0;
}
```



File Edit Selection View Go Run C++ practical

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS F:\Semester 2\C++\C++ practical> ./p2
Enter the no. of elements:3
Enter the elements:1
2
2
Array with unique elements:1 2
PS F:\Semester 2\C++\C++ practical> ./p2
Enter the no. of elements:5
Enter the elements:1
2
3
4
2
Array with unique elements:1 2 3 4
PS F:\Semester 2\C++\C++ practical> ./p2
Enter the no. of elements:5
Enter the elements:1
1
2
3
3
Array with unique elements:1 2 3
PS F:\Semester 2\C++\C++ practical>
PS F:\Semester 2\C++\C++ practical> ./p2
Enter the no. of elements:3
Enter the elements:1
1
1
Array with unique elements:1
PS F:\Semester 2\C++\C++ practical>
```

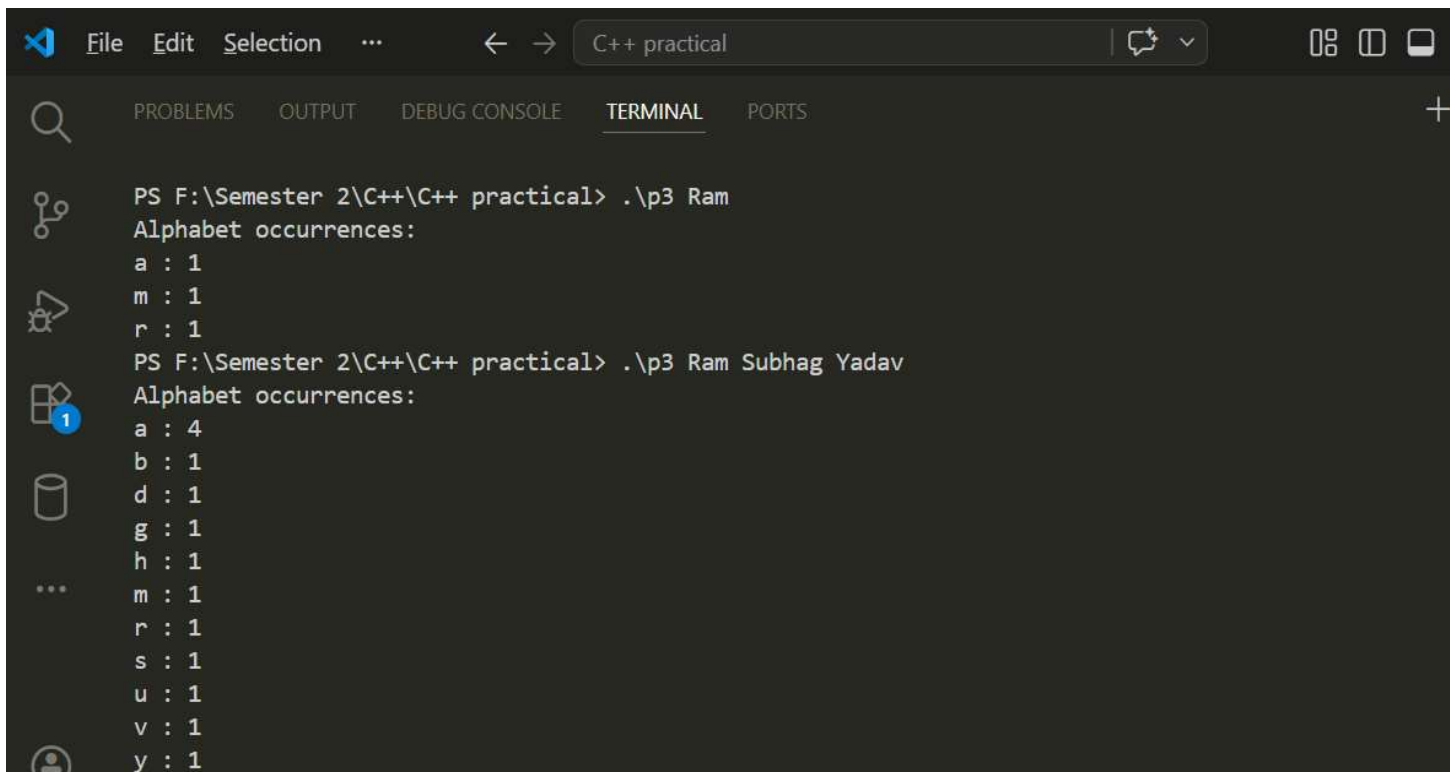
0 0 0 Connect BLACKBOX Agent Open Website Update is ready, click to restart. Ln 32, Col 2 Spaces: 4 UTF-8 CRLF {} C++ BLACKBOXAI: Open Ch

#(P3) Write a program that prints a table indicating the number of occurrences of each alphabet in the text entered as command line arguments.

```
#include <iostream>
#include <string>
#include <cctype>
using namespace std;

int main(int argc, char* argv[]) {
    int freq[26] = {0};
    for (int i = 1; i < argc; i++) {
        string text = argv[i];
        for (char ch : text) {
            if (isalpha(ch)){
                ch = tolower(ch);
                freq[ch - 'a']++;
            }
        }
    }
    cout << "Alphabet occurrences:\n";
    for (int i = 0; i < 26; i++) {
        if (freq[i] > 0) {
            cout << char('a' + i) << " : " << freq[i] << endl;
        }
    }

    return 0;
}
```



```
File Edit Selection ... C++ practical
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS F:\Semester 2\C++\C++ practical> .\p3 Ram
Alphabet occurrences:
a : 1
m : 1
r : 1
PS F:\Semester 2\C++\C++ practical> .\p3 Ram Subhag Yadav
Alphabet occurrences:
a : 4
b : 1
d : 1
g : 1
h : 1
m : 1
r : 1
s : 1
u : 1
v : 1
y : 1
```

""P(4): Write a menu driven program to perform string manipulation (without using inbuilt string functions):

- a. Show address of each character in string
- b. Concatenate two strings.
- c. Compare two strings
- d. Calculate length of the string (use pointers)
- e. Convert all lowercase characters to uppercase
- f. Reverse the string
- g. Insert a string in another string at a user specified position""

```
#include <iostream>
```

```
using namespace std;
```

```
int length(char *s) {  
    int len = 0;  
    while (*(s + len) != '\0') {  
        len++;  
    }  
    return len;  
}
```

```
void showAddress(char *s) {  
    for (int i = 0; s[i] != '\0'; i++) {  
        cout << "Character: " << s[i] << " Address: " << (void*)&s[i] << endl;  
    }  
}
```

```
void concatenate(char *s1, char *s2, char *result) {  
    int i = 0, j = 0;  
    while (s1[i] != '\0') {  
        result[i] = s1[i];  
        i++;  
    }  
    while (s2[j] != '\0') {  
        result[i] = s2[j];  
        i++; j++;  
    }  
    result[i] = '\0';  
}
```

```
int compare(char *s1, char *s2) {  
    int i = 0;  
    while (s1[i] != '\0' && s2[i] != '\0') {  
        if (s1[i] != s2[i]) {  
            return (s1[i] - s2[i]);  
        }  
        i++;  
    }  
    return (s1[i] - s2[i]);  
}
```

```
void toUppercase(char *s) {  
    for (int i = 0; s[i] != '\0'; i++) {  
        if (s[i] >= 'a' && s[i] <= 'z') {  
            s[i] = s[i] - 32;  
        }  
    }  
}
```

```
}  
}
```

```
void reverse(char *s) {  
    int len = length(s);  
    for (int i = 0; i < len / 2; i++) {  
        char temp = s[i];  
        s[i] = s[len - i - 1];  
        s[len - i - 1] = temp;  
    }  
}
```

```
void insertString(char *s1, char *s2, int pos, char *result) {  
    int i = 0, j = 0, k = 0;  
    int len1 = length(s1);  
    int len2 = length(s2);  
  
    for (i = 0; i < pos && i < len1; i++) {  
        result[k++] = s1[i];  
    }  
  
    for (j = 0; j < len2; j++) {  
        result[k++] = s2[j];  
    }  
  
    for (; i < len1; i++) {  
        result[k++] = s1[i];  
    }  
  
    result[k] = '\0';  
}
```

```
int main() {  
    char str1[100], str2[100], result[200];  
    int choice;  
  
    cout << "Enter a string: ";  
    cin.getline(str1, 100);  
  
    do {  
        cout << "\nMenu:\n";  
        cout << "1. Show address of each character\n";  
        cout << "2. Concatenate two strings\n";  
        cout << "3. Compare two strings\n";  
        cout << "4. Calculate length of the string\n";  
        cout << "5. Convert lowercase to uppercase\n";  
        cout << "6. Reverse the string\n";  
        cout << "7. Insert a string at position\n";  
        cout << "8. Exit\n";  
        cout << "Enter choice: ";  
        cin >> choice;
```



```

cin.ignore();

switch (choice) {
    case 1:
        showAddress(str1);
        break;
    case 2:
        cout << "Enter another string: ";
        cin.getline(str2, 100);
        concatenate(str1, str2, result);
        cout << "Concatenated string: " << result << endl;
        break;
    case 3:
        cout << "Enter another string: ";
        cin.getline(str2, 100);
        if (compare(str1, str2) == 0)
            cout << "Strings are equal\n";
        else
            cout << "Strings are not equal\n";
        break;
    case 4:
        cout << "Length of string: " << length(str1) << endl;
        break;
    case 5:
        toUppercase(str1);
        cout << "Uppercase string: " << str1 << endl;
        break;
    case 6:
        reverse(str1);
        cout << "Reversed string: " << str1 << endl;
        break;
    case 7:
        cout << "Enter another string: ";
        cin.getline(str2, 100);
        int pos;
        cout << "Enter position to insert: ";
        cin >> pos;
        insertString(str1, str2, pos, result);
        cout << "String after insertion: " << result << endl;
        break;
    case 8:
        cout << "Exiting program.\n";
        break;
    default:
        cout << "Invalid choice!\n";
}
} while (choice != 8);

return 0;
}

```

```
File Edit Selection View Go Run ... C++ practical
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS F:\Semester 2\C++\C++ practical> ./p4
Enter a string: Ram Subhag

Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Enter choice: 1
Character: R Address: 0x61feac
Character: a Address: 0x61fead
Character: m Address: 0x61feae
Character:   Address: 0x61feaf
Character: S Address: 0x61feb0
Character: u Address: 0x61feb1
Character: b Address: 0x61feb2
Character: h Address: 0x61feb3
Character: a Address: 0x61feb4
Character: g Address: 0x61feb5

Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Ln 56, Col 6 Spaces: 4 UTF-8 CRLF {} C++ BLACKBOXAI: Ope
```

```
File Edit Selection View Go ... C++ practical
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Enter choice: 2
Enter another string: Yadav
Concatenated string: Ram SubhagYadav

Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Enter choice: 4
Length of string: 10

Menu:
1. Show address of each character
Spaces: 4 UTF-8 CRLF {} C++ BLACKBOXAI: Open Cha
```

```
File Edit Selection View Go ... C++ practical
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Enter choice: 5
Uppercase string: RAM SUBHAG

Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Enter choice: 6
Reversed string: GAHBUS MAR

Menu:
1. Show address of each character
2. Concatenate two strings
```

```
File Edit Selection View Go Run ... C++ practical
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Exiting program.
PS F:\Semester 2\C++\C++ practical> ./p4
Enter a string: Ram Subhag

Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Enter choice: 7
Enter another string: Yadav
Enter position to insert: 10
String after insertion: Ram SubhagYadav

Menu:
1. Show address of each character
2. Concatenate two strings
3. Compare two strings
4. Calculate length of the string
5. Convert lowercase to uppercase
6. Reverse the string
7. Insert a string at position
8. Exit
Enter choice: 8
Exiting program.
PS F:\Semester 2\C++\C++ practical>
```

//P(5): Write a program to merge two ordered arrays to get a single ordered array.

```
#include <iostream>
using namespace std;

int main() {
    int n1, n2;

    cout << "Enter size of first array: ";
    cin >> n1;
    int arr1[n1];
    cout << "Enter elements of first array (sorted): ";
    for (int i = 0; i < n1; i++) {
        cin >> arr1[i];
    }

    cout << "Enter size of second array: ";
    cin >> n2;
    int arr2[n2];
    cout << "Enter elements of second array (sorted): ";
    for (int i = 0; i < n2; i++) {
        cin >> arr2[i];
    }

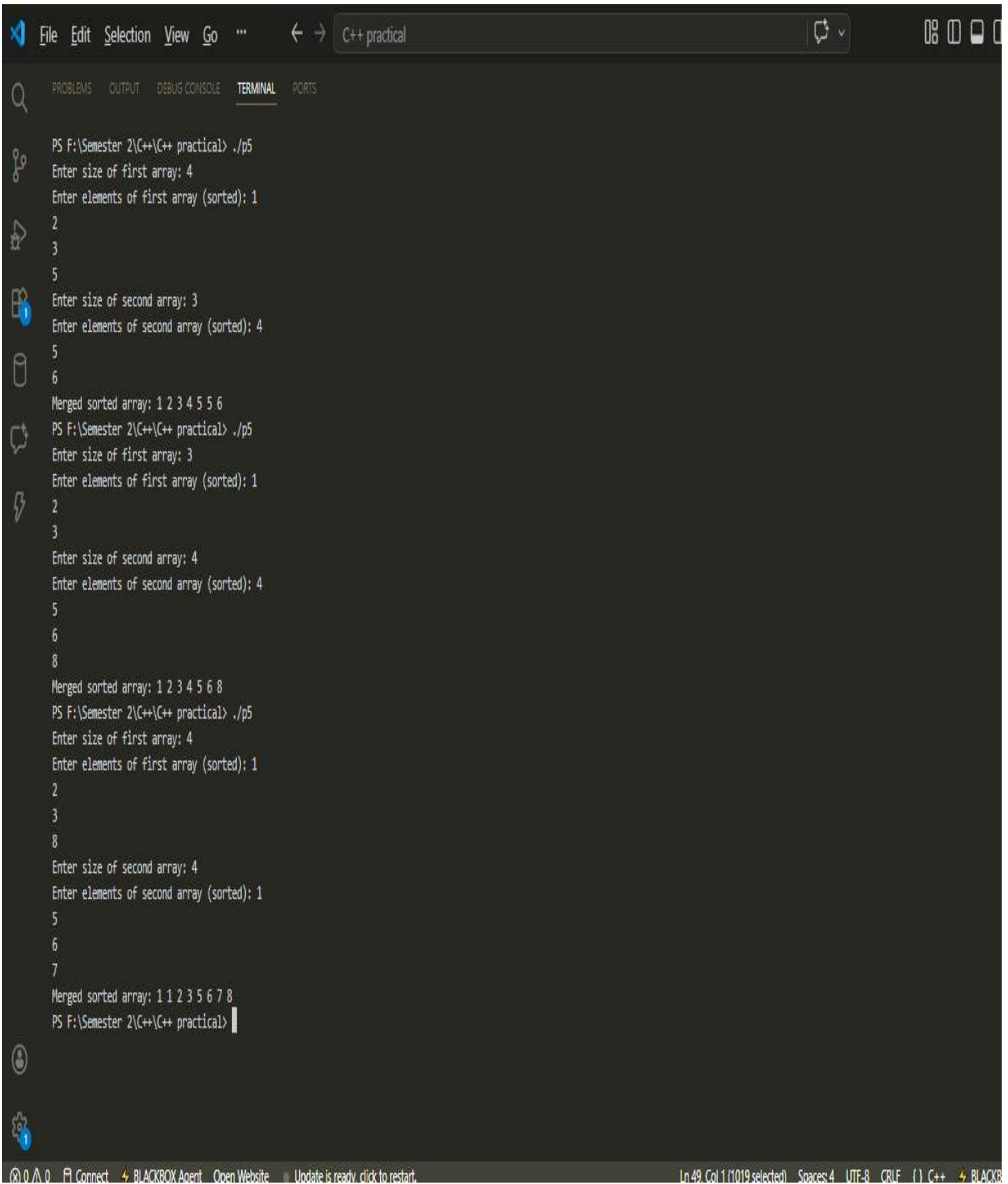
    int merged[n1 + n2];
    int i = 0, j = 0, k = 0;

    while (i < n1 && j < n2) {
        if (arr1[i] <= arr2[j]) {
            merged[k++] = arr1[i++];
        } else {
            merged[k++] = arr2[j++];
        }
    }

    while (i < n1) {
        merged[k++] = arr1[i++];
    }
    while (j < n2) {
        merged[k++] = arr2[j++];
    }

    cout << "Merged sorted array: ";
    for (int x = 0; x < n1 + n2; x++) {
        cout << merged[x] << " ";
    }
    cout << endl;

    return 0;
}
```



The image shows a Visual Studio Code editor window with a terminal pane open. The terminal displays the execution of a C++ program that merges two sorted arrays. The program is run three times with different input sizes. The first run uses arrays of size 4 and 3, resulting in a merged array of 1 2 3 4 5 6. The second run uses arrays of size 3 and 4, resulting in a merged array of 1 2 3 4 5 6 8. The third run uses arrays of size 4 and 1, resulting in a merged array of 1 1 2 3 5 6 7 8. The terminal output is as follows:

```
PS F:\Semester 2\C++\C++ practical> ./p5
Enter size of first array: 4
Enter elements of first array (sorted): 1
2
3
5
Enter size of second array: 3
Enter elements of second array (sorted): 4
5
6
Merged sorted array: 1 2 3 4 5 6
PS F:\Semester 2\C++\C++ practical> ./p5
Enter size of first array: 3
Enter elements of first array (sorted): 1
2
3
Enter size of second array: 4
Enter elements of second array (sorted): 4
5
6
8
Merged sorted array: 1 2 3 4 5 6 8
PS F:\Semester 2\C++\C++ practical> ./p5
Enter size of first array: 4
Enter elements of first array (sorted): 1
2
3
8
Enter size of second array: 4
Enter elements of second array (sorted): 1
5
6
7
Merged sorted array: 1 1 2 3 5 6 7 8
PS F:\Semester 2\C++\C++ practical>
```

//P(6): . Write a program to search a given element in a set of N numbers using Binary search

//(i)With recursion

```
#include <iostream>
using namespace std;

int binarySearch(int arr[], int low, int high, int key) {
    if (low > high) {
        return -1;
    }

    int mid = (low + high) / 2;

    if (arr[mid] == key) {
        return mid;
    }
    else if (key < arr[mid]) {
        return binarySearch(arr, low, mid - 1, key);
    }
    else {
        return binarySearch(arr, mid + 1, high, key);
    }
}

int main() {
    int n, key;
    cout << "Enter number of elements: ";
    cin >> n;

    int arr[n];
    cout << "Enter " << n << " sorted elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    cout << "Enter element to search: ";
    cin >> key;

    int result = binarySearch(arr, 0, n - 1, key);

    if (result != -1) {
        cout << "Element found at index " << result << endl;
    } else {
        cout << "Element not found in array." << endl;
    }

    return 0;
}
```

//(ii) Without recursion

```
#include <iostream>
using namespace std;

int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;

    while (low <= high) {
        int mid = (low + high) / 2;

        if (arr[mid] == key) {
            return mid;
        }
        else if (key < arr[mid]) {
            high = mid - 1;
        }
        else {
            low = mid + 1;
        }
    }

    return -1;
}

int main() {
    int n, key;
    cout << "Enter number of elements: ";
    cin >> n;

    int arr[n];
    cout << "Enter " << n << " sorted elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    cout << "Enter element to search: ";
    cin >> key;

    int result = binarySearch(arr, n, key);

    if (result != -1) {
        cout << "Element found at index " << result << endl;
    } else {
        cout << "Element not found in array." << endl;
    }

    return 0;
}
```

```
File Edit Selection View Go Run ... C++ practical 08 0
p7b.cpp text3.txt text4.txt p8.cpp p9.cpp p10.cpp p11.cpp p12.cpp p6b.cpp
p6b.cpp > main()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
compilation terminated.
PS F:\Semester 2\C++\C++ practical> g++ p6a.cpp -o p6a
PS F:\Semester 2\C++\C++ practical> ./p6a
Enter number of elements: 4
Enter 4 sorted elements: 1
4
5
6
Enter element to search: 5
Element found at index 2
PS F:\Semester 2\C++\C++ practical> ./p6a
Enter number of elements: 5
Enter 5 sorted elements: 1
2
4
5
7
Enter element to search: 7
Element found at index 4
PS F:\Semester 2\C++\C++ practical> ./p6a
Enter number of elements: 2
Enter 2 sorted elements: 1
7
Enter element to search: 1
Element found at index 0
PS F:\Semester 2\C++\C++ practical> 
```

```
File Edit Selection View Go Run ... C++ practical 08 0
p7b.cpp text3.txt text4.txt p8.cpp p9.cpp p10.cpp p11.cpp p12.cpp p6b.cpp
p6b.cpp > main()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
7
Enter element to search: 1
Element found at index 0
PS F:\Semester 2\C++\C++ practical> ./p6b
Enter number of elements: 4
Enter 4 sorted elements: 1
2
3
4
Enter element to search: 5
Element not found in array.
PS F:\Semester 2\C++\C++ practical> ./p6b
Enter number of elements: 5
Enter 5 sorted elements: 1
2
3
43
6
Enter element to search: 7
Element not found in array.
PS F:\Semester 2\C++\C++ practical> ./p6b
Enter number of elements: 3
Enter 3 sorted elements: 1
3
5
Enter element to search: 3
Element found at index 1
PS F:\Semester 2\C++\C++ practical> 
```


//P(7): Write a program to calculate GCD of two numbers

// (i) with recursion

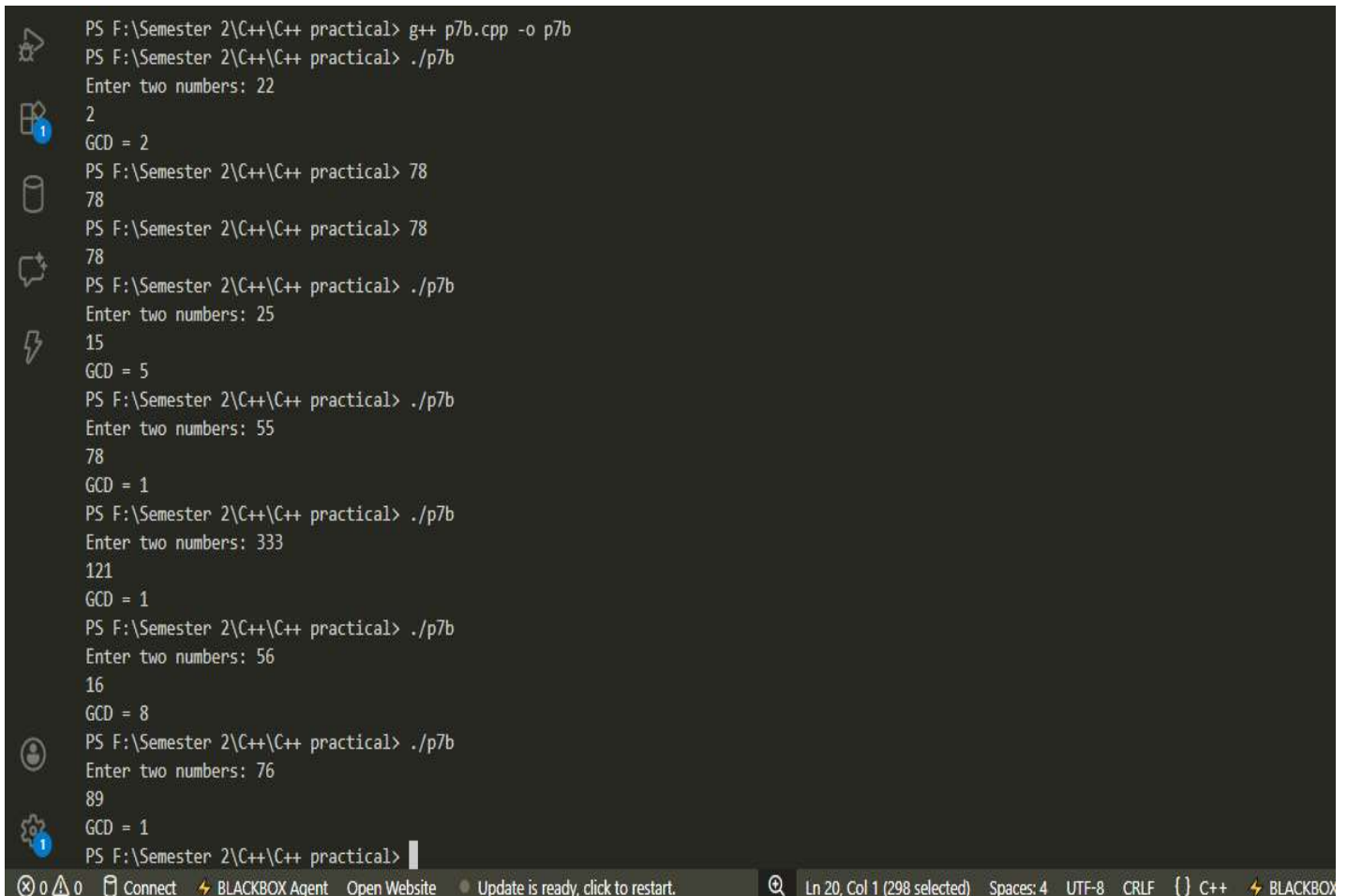
```
#include <iostream>
using namespace std;

int gcd(int m, int n) {
    if (n == 0) {
        return m;
    }
    return gcd(n, m % n);
}

int main() {
    int m, n;
    cout << "Enter two numbers: ";
    cin >> m >> n;

    cout << "GCD = " << gcd(m, n) << endl;

    return 0;
}
```



The screenshot shows a Windows Command Prompt window with the following text:

```
PS F:\Semester 2\C++\C++ practical> g++ p7b.cpp -o p7b
PS F:\Semester 2\C++\C++ practical> ./p7b
Enter two numbers: 22
2
GCD = 2
PS F:\Semester 2\C++\C++ practical> 78
78
PS F:\Semester 2\C++\C++ practical> 78
78
PS F:\Semester 2\C++\C++ practical> ./p7b
Enter two numbers: 25
15
GCD = 5
PS F:\Semester 2\C++\C++ practical> ./p7b
Enter two numbers: 55
78
GCD = 1
PS F:\Semester 2\C++\C++ practical> ./p7b
Enter two numbers: 333
121
GCD = 1
PS F:\Semester 2\C++\C++ practical> ./p7b
Enter two numbers: 56
16
GCD = 8
PS F:\Semester 2\C++\C++ practical> ./p7b
Enter two numbers: 76
89
GCD = 1
PS F:\Semester 2\C++\C++ practical>
```

The status bar at the bottom of the window shows: 0 0 0, Connect, BLACKBOX Agent, Open Website, Update is ready, click to restart, Ln 20, Col 1 (298 selected), Spaces: 4, UTF-8, CRLF, {}, C++, and BLACKBOX.

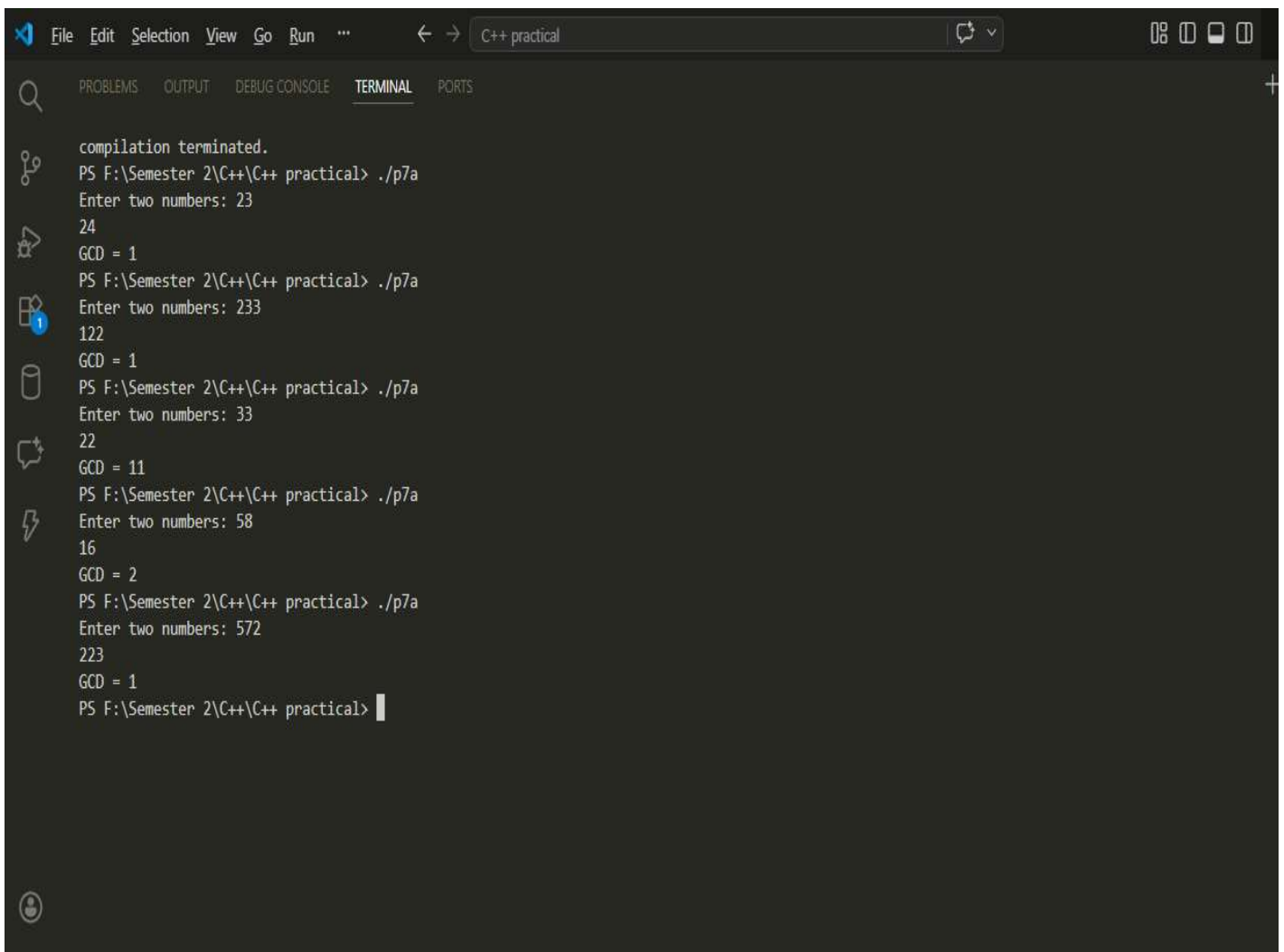
//(ii) Without recursion

```
#include <iostream>
using namespace std;

int main() {
    int m, n;
    cout << "Enter two numbers: ";
    cin >> m >> n;

    while (n != 0) {
        int temp = n;
        n = m % n;
        m = temp;
    }

    cout << "GCD = " << m << endl;
    return 0;
}
```



The screenshot shows a Visual Studio Code interface with a terminal window open. The terminal displays the output of a C++ program that calculates the Greatest Common Divisor (GCD) of two numbers using a while loop. The program prompts the user to enter two numbers, and the terminal shows several test cases with their corresponding GCD results.

```
compilation terminated.
PS F:\Semester 2\C++\C++ practical> ./p7a
Enter two numbers: 23
24
GCD = 1
PS F:\Semester 2\C++\C++ practical> ./p7a
Enter two numbers: 233
122
GCD = 1
PS F:\Semester 2\C++\C++ practical> ./p7a
Enter two numbers: 33
22
GCD = 11
PS F:\Semester 2\C++\C++ practical> ./p7a
Enter two numbers: 58
16
GCD = 2
PS F:\Semester 2\C++\C++ practical> ./p7a
Enter two numbers: 572
223
GCD = 1
PS F:\Semester 2\C++\C++ practical> 
```

""P(8): . Create a Matrix class. Write a menu-driven program to perform following Matrix operations (exceptions should be thrown by the functions if matrices passed to them are incompatible and handled by the main() function):

- a. Sum
- b. Product
- c. Transpose""

```
#include <iostream>
using namespace std;
```

```
class Matrix {
```

```
private:
```

```
    int rows, cols;
```

```
    int mat[10][10];
```

```
public:
```

```
    Matrix(int r = 0, int c = 0) {
```

```
        rows = r;
```

```
        cols = c;
```

```
    }
```

```
    void input() {
```

```
        cout << "Enter elements (" << rows << "x" << cols << "):\n";
```

```
        for (int i = 0; i < rows; i++) {
```

```
            for (int j = 0; j < cols; j++) {
```

```
                cin >> mat[i][j];
```

```
            }
```

```
        }
```

```
    }
```

```
    void display() {
```

```
        for (int i = 0; i < rows; i++) {
```

```
            for (int j = 0; j < cols; j++) {
```

```
                cout << mat[i][j] << " ";
```

```
            }
```

```
            cout << endl;
```

```
        }
```

```
    }
```

```
    Matrix add(Matrix other, bool &ok) {
```

```
        Matrix result;
```

```
        if (rows != other.rows || cols != other.cols) {
```

```
            ok = false;
```

```
            return result;
```

```
        }
```

```
        ok = true;
```

```
        result.rows = rows;
```

```
        result.cols = cols;
```

```
        for (int i = 0; i < rows; i++) {
```

```

        for (int j = 0; j < cols; j++) {
            result.mat[i][j] = mat[i][j] + other.mat[i][j];
        }
    }
    return result;
}

```

```

Matrix multiply(Matrix other, bool &ok) {
    Matrix result;
    if (cols != other.rows) {
        ok = false;
        return result;
    }
    ok = true;
    result.rows = rows;
    result.cols = other.cols;
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < other.cols; j++) {
            result.mat[i][j] = 0;
            for (int k = 0; k < cols; k++) {
                result.mat[i][j] += mat[i][k] * other.mat[k][j];
            }
        }
    }
    return result;
}

```

```

Matrix transpose() {
    Matrix result(cols, rows);
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            result.mat[j][i] = mat[i][j];
        }
    }
    return result;
}

```

```

int getRows() { return rows; }
int getCols() { return cols; }
};

```

```

int main() {
    int r1, c1, r2, c2;
    cout << "Enter rows and cols of Matrix A: ";
    cin >> r1 >> c1;
    Matrix A(r1, c1);
    A.input();

    cout << "Enter rows and cols of Matrix B: ";
    cin >> r2 >> c2;
    Matrix B(r2, c2);
}

```

```

B.input();

int choice;
do {
    cout << "\nMenu:\n";
    cout << "1. Sum\n";
    cout << "2. Product\n";
    cout << "3. Transpose (Matrix A)\n";
    cout << "4. Exit\n";
    cout << "Enter choice: ";
    cin >> choice;

    bool ok;
    switch (choice) {
        case 1: {
            Matrix C = A.add(B, ok);
            if (ok) {
                cout << "Sum:\n";
                C.display();
            } else {
                cout << "Error: Matrices incompatible for addition!\n";
            }
            break;
        }
        case 2: {
            Matrix C = A.multiply(B, ok);
            if (ok) {
                cout << "Product:\n";
                C.display();
            } else {
                cout << "Error: Matrices incompatible for multiplication!\n";
            }
            break;
        }
        case 3: {
            Matrix C = A.transpose();
            cout << "Transpose of Matrix A:\n";
            C.display();
            break;
        }
        case 4:
            cout << "Exiting program.\n";
            break;
        default:
            cout << "Invalid choice!\n";
    }
} while (choice != 4);

return 0;
}

```

```
File Edit Selection View Go ... C++ practical
Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSReadLine'

PS F:\Semester 2\C++\C++ practical> .\p8
Enter rows and cols of Matrix A: 3 3
Enter elements (3x3):
1 2 3
3 4 2
5 6 4
Enter rows and cols of Matrix B: 3 3
Enter elements (3x3):
1 2 8
4 5 7
3 4 5

Menu:
1. Sum
2. Product
3. Transpose (Matrix A)
4. Exit
Enter choice: 1
Sum:
2 4 11
7 9 9
8 10 9

Menu:
1. Sum
2. Product
3. Transpose (Matrix A)
4. Exit
Enter choice: 2
Product:
18 24 37
25 34 62
41 56 102

Menu:
1. Sum
2. Product
3. Transpose (Matrix A)
4. Exit
Enter choice: 3
Transpose of Matrix A:
1 3 5
2 4 6
3 2 4

Menu:
1. Sum
2. Product
3. Transpose (Matrix A)
4. Exit
Enter choice: 4
Exiting program.
PS F:\Semester 2\C++\C++ practical>
```

```
File Edit Selection View Go Run ... C++ practical
8 10 9

Menu:
1. Sum
2. Product
3. Transpose (Matrix A)
4. Exit
Enter choice: 2
Product:
18 24 37
25 34 62
41 56 102

Menu:
1. Sum
2. Product
3. Transpose (Matrix A)
4. Exit
Enter choice: 3
Transpose of Matrix A:
1 3 5
2 4 6
3 2 4

Menu:
1. Sum
2. Product
3. Transpose (Matrix A)
4. Exit
Enter choice: 4
Exiting program.
PS F:\Semester 2\C++\C++ practical>
```

""P(9) Define a class Person having name as a data member. Inherit two classes Student and Employee from Person. Student has additional attributes as course, marks and year and Employee has department and salary. Write display() method in all the three classes to display the corresponding attributes. Provide the necessary methods to show runtime polymorphism.""

```
#include <iostream>
#include <string>
using namespace std;
```

```
class Person {
protected:
    string name;

public:
    Person(string n = "") : name(n) {}

    virtual void display() {
        cout << "Name: " << name << endl;
    }
};
```

```
class Student : public Person {
private:
    string course;
    int marks;
    int year;

public:
    Student(string n, string c, int m, int y) : Person(n), course(c), marks(m), year(y) {}

    void display() override {
        cout << "Student Name: " << name << endl;
        cout << "Course: " << course << endl;
        cout << "Marks: " << marks << endl;
        cout << "Year: " << year << endl;
    }
};
```

```
class Employee : public Person {
private:
    string department;
    double salary;

public:
    Employee(string n, string d, double s) : Person(n), department(d), salary(s) {}

    void display() override {
        cout << "Employee Name: " << name << endl;
        cout << "Department: " << department << endl;
        cout << "Salary: " << salary << endl;
    }
};
```

```

    }
};

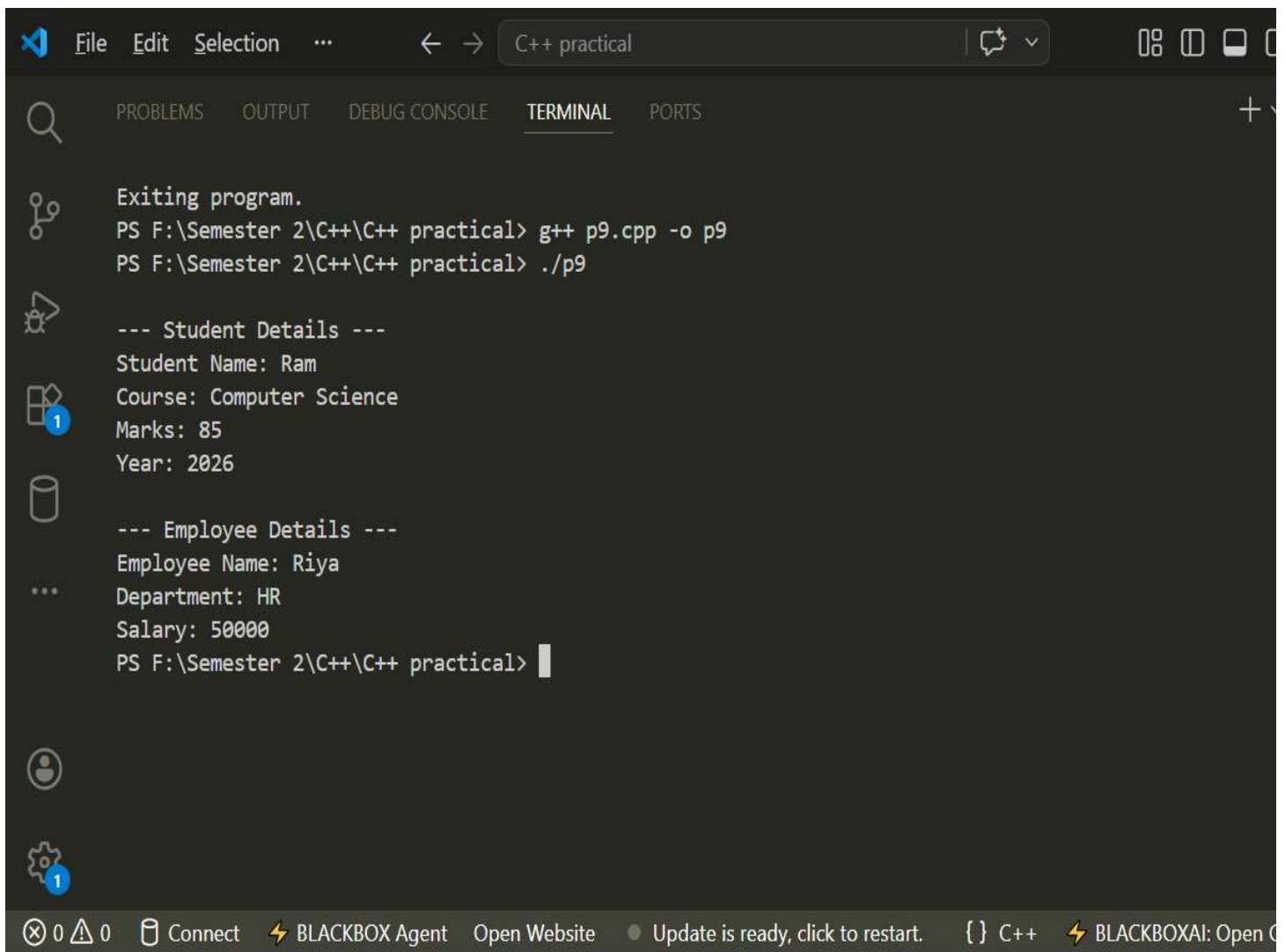
int main() {
    Person* p;

    Student s("Ram", "Computer Science", 85, 2026);
    p = &s;
    cout << "\n--- Student Details ---\n";
    p->display();

    Employee e("Riya", "HR", 50000);
    p = &e;
    cout << "\n--- Employee Details ---\n";
    p->display();

    return 0;
}

```



The screenshot shows a Visual Studio Code interface with a terminal window open. The terminal title is "C++ practical". The output of the program is as follows:

```

Exiting program.
PS F:\Semester 2\C++\C++ practical> g++ p9.cpp -o p9
PS F:\Semester 2\C++\C++ practical> ./p9

--- Student Details ---
Student Name: Ram
Course: Computer Science
Marks: 85
Year: 2026

--- Employee Details ---
Employee Name: Riya
Department: HR
Salary: 50000
PS F:\Semester 2\C++\C++ practical>

```

The bottom status bar of the IDE shows icons for search, connect, BLACKBOX Agent, Open Website, Update is ready, click to restart, C++ language mode, and BLACKBOXAI: Open Chat.

""P(10) Create a Triangle class. Add exception handling statements to ensure the following conditions: all sides are greater than 0 and sum of any two sides are greater than the third side. The class should also have overloaded functions for calculating the area of a right angled triangle as well as using Heron's formula to calculate the area of any type of triangle""

```
#include <iostream>
#include <cmath>
using namespace std;

class Triangle {
private:
    double a, b, c;

public:

    Triangle(double x, double y, double z) {
        if (x <= 0 || y <= 0 || z <= 0) {
            throw "All sides must be greater than 0!";
        }
        if (x + y <= z || x + z <= y || y + z <= x) {
            throw "Sum of any two sides must be greater than the third side!";
        }
        a = x; b = y; c = z;
    }

    double area(double base, double height) {
        if (base <= 0 || height <= 0) {
            throw "Base and height must be positive!";
        }
        return 0.5 * base * height;
    }

    double area() {
        double s = (a + b + c) / 2.0;
        return sqrt(s * (s - a) * (s - b) * (s - c));
    }

    void displaySides() {
        cout << "Sides: " << a << ", " << b << ", " << c << endl;
    }
};

int main() {
    try {
        double x, y, z;
        cout << "Enter three sides of the triangle: ";
        cin >> x >> y >> z;

        Triangle t(x, y, z);
        t.displaySides();
    }
}
```

```

cout << "Area using Heron's formula: " << t.area() << endl;

double base, height;
cout << "\nEnter base and height for right-angled triangle: ";
cin >> base >> height;
cout << "Area of right-angled triangle: " << t.area(base, height) << endl;
}
catch (const char* msg) {
    cout << "Error: " << msg << endl;
}

return 0;
}

```

The screenshot shows a Visual Studio Code terminal window with the title bar "C++ practical". The terminal output is as follows:

```

compilation terminated.
PS F:\Semester 2\C++\C++ practical> g++ p10.cpp -o p10
PS F:\Semester 2\C++\C++ practical> ./p10
Enter three sides of the triangle: 3 4 5
Sides: 3, 4, 5
Area using Heron's formula: 6

Enter base and height for right-angled triangle: 3 4
Area of right-angled triangle: 6
PS F:\Semester 2\C++\C++ practical> ./p10
Enter three sides of the triangle: 4 7 8
Sides: 4, 7, 8
Area using Heron's formula: 13.9978

Enter base and height for right-angled triangle: 4 5
Area of right-angled triangle: 10
PS F:\Semester 2\C++\C++ practical> ./p10
Enter three sides of the triangle: 3 6 9
Error: Sum of any two sides must be greater than the third side!
PS F:\Semester 2\C++\C++ practical>

```

The terminal window includes a sidebar on the left with icons for Explorer, Search, Run and Debug, and Extensions. The bottom status bar shows "0" errors, "Connect" button, "BLACKBOX Agent", "Open Website", "Update is ready, click to restart.", "C++", and "BLACKBOXAI: Open Chat".

""P(11) Create a class Student containing fields for Roll No., Name, Class, Year and Total Marks. Write a program to store 5 objects of Student class in a file. Retrieve these records from the file and display them.

```
#include <iostream>
#include <fstream>
using namespace std;

class Student {
private:
    int rollNo;
    string name;
    string studentClass;
    int year;
    double totalMarks;

public:
    void input() {
        cout << "Enter Roll No: ";
        cin >> rollNo;
        cin.ignore(); // clear newline
        cout << "Enter Name: ";
        getline(cin, name);
        cout << "Enter Class: ";
        getline(cin, studentClass);
        cout << "Enter Year: ";
        cin >> year;
        cout << "Enter Total Marks: ";
        cin >> totalMarks;
    }

    void writeToFile(ofstream &fout) {
        fout << rollNo << "," << name << "," << studentClass << ","
            << year << "," << totalMarks << endl;
    }

    void readFromFile(string line) {
        string values[5];
        int field = 0;
        string temp = "";

        for (char ch : line) {
            if (ch == ',') {
                values[field++] = temp;
                temp = "";
            } else {
                temp += ch;
            }
        }
    }
}
```

```

    values[field] = temp;

    rollNo = stoi(values[0]);
    name = values[1];
    studentClass = values[2];
    year = stoi(values[3]);
    totalMarks = stod(values[4]);
}

void display() {
    cout << "Roll No: " << rollNo
        << ", Name: " << name
        << ", Class: " << studentClass
        << ", Year: " << year
        << ", Total Marks: " << totalMarks << endl;
}
};

int main() {
    Student students[5];

    ofstream fout("students.txt");
    if (!fout) {
        cout << "Error opening file for writing!" << endl;
        return 1;
    }

    cout << "Enter details of 5 students:\n";
    for (int i = 0; i < 5; i++) {
        cout << "\nStudent " << i + 1 << ": \n";
        students[i].input();
        students[i].writeToFile(fout);
    }
    fout.close();

    ifstream fin("students.txt");
    if (!fin) {
        cout << "Error opening file for reading!" << endl;
        return 1;
    }

    cout << "\n--- Student Records Retrieved from File ---\n";
    string line;
    while (getline(fin, line)) {
        Student s;
        s.readFromFile(line);
        s.display();
    }
    fin.close();

    return 0;
}

```

```
File Edit Selection View Go Run ... C++ practical
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS F:\Semester 2\C++\C++ practical> ./p11
Enter details of 5 students:

Student 1:
Enter Roll No: 101
Enter Name: ram
Enter Class: 3
Enter Year: 1
Enter Total Marks: 203

Student 2:
Enter Roll No: 102
Enter Name: Subhag
Enter Class: 4
Enter Year: 1
Enter Total Marks: 1

Student 3:
Enter Roll No: 103
Enter Name: Kishor
Enter Class: 5
Enter Year: 2021
Enter Total Marks: 309

Student 4:
Enter Roll No: 104
Enter Name: Dipak
Enter Class: 6
Enter Year: 3
Enter Total Marks: 300

Student 5:
Enter Roll No: 105
Enter Name: Raju
Enter Class: 5
Enter Year: 3
Enter Total Marks: 400

Ln 99, Col 2 (2333 selected) Spaces: 4 UTF-8 CRLF {} C++ BLACKBOXAI: Oper
```

```
File Edit Selection View Go Run ... C++ practical
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Student 5:
Enter Roll No: 105
Enter Name: Raju
Enter Class: 5
Enter Year: 3
Enter Total Marks: 400

--- Student Records Retrieved from File ---
Roll No: 101, Name: ram, Class: 3, Year: 1, Total Marks: 203
Roll No: 102, Name: Subhag, Class: 4, Year: 1, Total Marks: 1
Roll No: 103, Name: Kishor, Class: 5, Year: 2021, Total Marks: 309
Roll No: 104, Name: Dipak, Class: 6, Year: 3, Total Marks: 300
Roll No: 105, Name: Raju, Class: 5, Year: 3, Total Marks: 400
PS F:\Semester 2\C++\C++ practical>
```

//P(12) Copy the contents of one text file to another file, after removing all whitespaces.

```
#include <fstream>
using namespace std;
int main(){
    ifstream fin;
    ofstream fout;
    fin.open("text3.txt");
    fout.open("text4.txt");
    char ch;
    while(fin.get(ch)){
        if (ch==' '){
            continue;
        }
        else
            fout<<ch;
    }
    return 0;
}
```

